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Data Structure Analysis

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The Ethics of Automatic Software Testing

Automated unit testing is a form of programming where a developer makes small tests to his or her program to validate the ability of their program to function as intended. While developers can use this technique to ensure the validity of their program while on a computer, it can still run into faults when used in the real world. The case of the Boeing 747 MAX is an extremely relevant example of what could happen when the software passes the given tests but can still fail when put in a real situation with uncontrollable outside factors. In this way, it is extremely important for a developer to consider any outside factors on their program that could cause it to fail, as it could potentially lead to things like the Boeing 747 MAX incident. From a Christian worldview, it is our responsibility to care for others and to be an example for other people for what the love of God looks like, including in development and programming.

The 2019 incident involving the Boeing 747 MAX is an example of what can happen when unit testing is not done properly or is not understood at a fundamental level. The testing done was not conducted under real world conditions, and as such, the plane was not prepared for certain variables out of human control and tragically crashed. Rather than testing all cases, the developers presumed that their program would function as intended under all edge cases, however, they were incorrect. This highlights one of the main ethical concerns of unit testing,

which is properly determining if your code behaves as intended under all possible circumstances, no matter how unlikely.

The ACM code of ethics is a great tool to analyze ethical issues within computer science. The code emphasizes avoiding harm to other people, and complete transparency about the potential shortcomings of your program (ACM). Avoiding certain edge cases to satisfy test completion would naturally violate the set rules from the ACM. Because of this failure, Boeing went under investigation, and many forms of negligence were found in the creation of the 737, and it was grounded as a result (Transportation and Infrastructure). When developers disregard ethical codes such as the ACM, they risk the safety and wellbeing of their users.

A Christian worldview furthers the responsibility that a developer holds with any ethical issue. As a Christian, it is my calling to care for others, and to love others. The book of 1 Peter says, “Above all, love each other deeply, because love covers over a multitude of sins.” This command extends to everyone, including those who will be using the programs that I will write one day. Treating those people with love means to ensure that every piece of code I write is safe, functional, and well tested. Not properly testing my code is also not treating those who will be using it with love and is in direct defiance to how the Bible calls me to live.

Beyond my own personal convictions, the Bible also calls people to use their talents and skills in a way that is glorifying to God. Developers have been given strong skill sets and knowledge within their field, and the Bible makes it very clear they should use their talents for the benefit of others. 1 Peter goes on to say, “Each of you should use whatever gift you have received to serve others, as faithful stewards of God’s grace in its various forms.” Developers need to be honest with their users about the limitations of their code, especially when the software they create will directly impact other human lives.

Ethical issues like that of the Boeing 737's unit testing are proof that unit testing should be at the forefront of a developer's mind. Unit testing should not be just a way for a developer to check off boxes, but rather a way for them to operate ethically and ensure the absolute safety of the users of their software. A Christian centered worldview calls a developer to view their users with love and prioritize their safety and prevent harm. Therefore, developers should treat unit testing with the mindset of protecting other people and keeping their users safe.

Works Cited

ACM Code of Ethics and Professional Conduct, www.acm.org/code-of-ethics. Accessed 6 Feb. 2026.

"Boeing 737 MAX Investigation: The House Committee on Transportation and Infrastructure." *Boeing 737 MAX Investigation*, The House Committee on Transportation and Infrastructure, democrats-transportation.house.gov/committee-activity/boeing-737-max-investigation. Accessed 5 Feb. 2026.